

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian	Molecules Electroporated	DNA: pRC/CMV- Human 5-HT(10β) receptor construct, linearized. [See notes]
Species Used	Human: HeLa, epithelial carcinoma		

Before the Pulse

Cell Growth Medium	EMEM + 10 % Fetal Calf Serum, + Non-essential amino acids + Pen / Strep	Growth Phase at Harvest	Pre-confluent
		Pre-pulse Incubation	10 min on ice

Wash Solution Phosphate Buffered Sucrose

The Pulse

Instruments Used Gene Pulser® apparatus & Pulse Controller

Electroporation Temperature	about 4 °C		
Electroporation Medium*	Phosphate Buffer Sucrose (see Gene Pulser Instruction Manual)	Cuvette Gap	0.4 cm
Cell Density	5 x 10 (6) cells / 800 µl	Voltage	0.4 kV
Volume of Cells	400 µl	Field Strength	1.0 kV/cm
DNA Concentration	20 µg / pulse		
DNA Resuspension Buffer	Phosphate Buffered Sucrose	Capacitor	25 µF
Volume of DNA	400 µl	Resistor	(Pulse Controller) 200 Ω
		Time Constant	2.5 msec

After the Pulse

Outgrowth Medium Post pulse: (in cuvette) 10 min. on ice, then
EMEM + 10 % FCS,
+ non-essential a.a.+ Pen / Strep

Outgrowth Temperature	37 °C
Length of Incubation	plated, split next day
Selection Method or Assay Used	G418 selection (800 µg/ml) 24 hours post pulse
Electroporation Efficiency	100 transformants / µg
Per Cent Survival	Not known

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

DNA: 7.2 kB linear ds DNA construct including 5.6 kB of the mammalian expression vector pRC/ CMV (Invitrogen) and 1.8 kB of human genomic DNA including the coding region of the 5-HT10b (5 HT-1B) serotonin receptor, linearized with *Bgl* II.

Phosphate Buffered Sucrose: 272 mM sucrose, 7 mM sodium phosphate, pH 7.4, 1 mM MgCl₂.

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196